**Referee report on JEDC-D-26-00041**

**Sorting, Price Informativeness, and Information Acquisition in Dark Markets**

The authors build a laboratory experiment to investigate the impact of introducing a dark pool on information acquisition, venue choice, and market outcomes such as price discovery and liquidity. The paper is very well written, and the design of the experiment is clever – especially the modelling of information. There is much to like about the experiment, although I have some concerns about the structure of the paper and the mechanisms involved.

## Concerns

**Item 1**: **Contribution.** The most interesting contribution of the study is, in my view, related to information acquisition and price discovery. I believe this should be made front and center and investigated further in depth. Currently, the paper opens with results on substitutability of venues (which is somewhat trivial) and liquidity – which harks back to Bloomfield, O'Hara, and Saar (2015) and is not particularly suprising. However, the impact of dark trading on price discovery *through information acquisition* is a fresh idea and a genuine novel contribution.

I believe the most natural causal chain is information acquisition, venue choice and order submission, price discovery and liquidity – which is the opposite order as to how results are currently presented in the paper.

**Item 2**: **Information acquisition.** The exciting result of the paper, in my view, is that introducing a dark pool leads to enhanced information production. However, at this stage, the channel is not sufficiently explored in depth: the authors propose some hypotheses, but cannot test them. I am intrigued by the fact that the result goes against the model of Zhu (2013): it would be a powerful contribution to explain exactly how that theoretical framework may fail to explain reality.

I am inclined to believe that ex ante, the promise of trading anonymously leads to participants wanting to become more informed. However, with experience, seeing that their execution odds are better in lit markets, the effect should vanish. Table 7 does show that, indeed, the information acquisition effect diminishes with experience, but it does not seem to explain the full story.

I would perhaps elicit beliefs from traders on how many other signals they think will be acquired in a given round, or where do they expect to trade? For example, traders might believe that with dark trading enabled, others will become more informed so they themselves must do so as well even if they plan to trade on lit markets (level-*k* reasoning).

At the same time, I am puzzled by the fact that even though traders acquire more information, and informed traders choose lit markets to avoid execution risk, price discovery does not improve in a statistically significant way. Why is this?

Generally, I would encourage the authors to map the results and the economics closely to Zhu (2013) and point out exactly where the model aligns or not, and what does the experiment say about the underlying assumptions that might be violated? One important difference, for example, is that Zhu (2013) does not have endogenous information acquisition.

The effect of dark trading on volume and spreads is, in my view, a secondary result which can safely be relegated to a "sanity check" section or even the appendix.

**Item 3**: **Treatments.** First, as a note, I am impressed by the stark difference in results between the low- and the high-information environments. I would have expected to see the same effects in the low-information sessions, albeit weaker.

However, given the sheer amount of null results in those treatments, I am not sure we learn so much from low-information sessions. I was wondering if the authors can perhaps replace them with two additional sessions where information is exogenous: some participants receive signals, others do not. That model would be, by and large, a replication of Zhu (2013).

I believe this treatment structure should help us understand how much of the effect of dark trading on market quality comes from changes in information acquisition, and how much comes directly from market structure.

**Item 4**: **Writing.** I believe some of the terminology could be tighter. “Sorting” is essentially “venue choice” in this context, for example, and the latter is perhaps easier to grasp for microstructure researchers, who are an important audience for this paper.

**Item 5**: **Value of signals.** It would be interesting to know how the value of 6 ECU for the signal was calibrated – was it through pilot studies or a decision model?

**Item 6**: **Empirical results.** Generally, I would encourage you to standardize covariates such that the constants are better interpretable as average effects for some group. For the point above, this would have helped me understand how many signals people acquire on average. For the spreads in Table 2 and 3, the constant term is different by an order of magnitude, which muddles interpretation for the effect of dark markets (and is puzzling in and of itself).

## References


Bloomfield, Robert, Maureen O’Hara, and Gideon Saar, 2015, Hidden Liquidity: Some New Light on Dark Trading, *Journal of Finance* 70, 2227–2274.

Zhu, Haoxiang, 2013, Do Dark Pools Harm Price Discovery?, *The Review of Financial Studies* 27, 747–789.